

Ahmed Abdel Latif Abdel Aziz Salem

Freelance Software & AI Developer | Computer Vision | Full-Stack

[City, Egypt] • [+20 1212817383] • ahmedzizu81@gmail.com

github.com/Ahmedzizu • linkedin.com/in/ahmed-abdel-latif-a893992b1

PROFILE

Computer Science student (AI track, Benha University) working as a freelance developer for real clients. I analyze a client's manual operations, design the system that digitizes them, and deliver it end-to-end — from requirements to production. Delivered an award-winning AI railway inspection system (YOLOv8, mAP@0.5 = 0.94), a multi-property hotel management platform, and an enterprise training-center SaaS. Comfortable across the whole stack: ESP32 hardware and circuits, Computer Vision model training on custom datasets, and React/Node.js platforms. I use AI-assisted development tools daily to move faster without sacrificing quality.

EXPERIENCE

Freelance Software & AI Developer — Self-employed 2025 – Present

- Work directly with business clients: gather requirements, analyze existing manual workflows, and translate them into working software — owning planning, architecture, and implementation together with one development partner.
- Delivered the full technical implementation of an AI railway inspection system for a graduation-project team at 6 October University — a client who found me through my earlier robotics project “Qader AI”; the engagement is still ongoing.
- Built and shipped two management platforms (hotel management and training-center management) described in Projects below — both replacing fully manual operations.
- Use AI-assisted development tools (LLM pair-programming) as a core part of the workflow to accelerate delivery and code review.

PROJECTS

Railway Defect Detection System — YOLOv8 + ESP32 (freelance, award-winning) 2026

- **1st place (college) & 2nd place (university) — 6 October University graduation projects competition.**
- Dual-camera inspection design: front camera analyzes the track before the train passes, rear camera re-checks after — catching defects caused by the passage itself.
- Built a custom dataset from scratch (no public dataset existed): manual annotation of 6 defect classes, augmentation for day/night lighting, vibration, and noise — 2,769 images.
- Diagnosed and fixed real data problems: mixed bounding-box/polygon annotations and severe class imbalance (4,457 vs 604 instances); trained 10+ model iterations.
- Final model (YOLOv8s, 100 epochs, AdamW): mAP@0.5 = 0.94, Precision = 0.92, Recall = 0.93. Exported to ONNX and benchmarked PT vs ONNX inference on real video with OpenCV.
- Also owned the hardware side: component selection, power calculation, circuit design (Fritzing), and ESP32/ESP32-CAM programming.
- Code, weights, and training results: github.com/Ahmedzizu/rail-defect-detection

Sidre — Multi-Property Hotel Management Platform (freelance) 2025

- Web platform unifying the management of 5 hotels/chalets in one system for the owner — replacing a fully manual operation. Planned and built end-to-end with one partner, starting from on-site requirements analysis with the client.
- Reservations management, financial reporting, and a cash-treasury module with cash-shift tracking for front-desk staff.

- Role-based access control: employee permissions determine which hotels' reservations and finances each user can see.
- Built a WhatsApp bot integration: every reservation action (approval, modification, cancellation) automatically notifies the client with the booking status.
- Stack: React, Node.js/Express, MongoDB.

Riyadat — Training Center Management SaaS (freelance)

2025 – Present

- Enterprise system for a training institute: companies, trainees, programs, groups, scheduling, attendance, and reporting — planned and implemented end-to-end with one partner from client requirements.
- Full Arabic/English RTL support (i18next); React 18 + Vite frontend, Node.js/Express + PostgreSQL backend with role-based permissions and audit logging.
- Integrated ZKTeco fingerprint attendance devices into the platform; built attendance operations screens and a reports module with charts, insights, and Arabic-compatible CSV export.

Qader A1 — Autonomous Robotic System (2 versions)

2024

- Co-developed a two-person robotics project; owned the hardware: component selection, electronics, sensors, and assembly (partner owned software).
- V1: Arduino Uno + IR control + ultrasonic distance logic. V2: ESP32 serving its own web control interface over a Wi-Fi Access Point, with laser targeting and return-to-start using stored telemetry.
- This project is what brought in the railway-inspection freelance client. Repo: github.com/0xmaro/Qader-A1

Line-Follower Vehicle — pure logic gates (Logic Design course)

2023

- Built a black-line-following vehicle with no microcontroller and no code: IR sensors driving the motors through a control circuit made entirely of logic gates.

Telco Customer Churn Prediction — NTI × Huawei (HCIA-AI track graduation project)

2025

- ML pipeline: preprocessing, EDA, feature engineering; trained Logistic Regression, Random Forest, and XGBoost; evaluated with precision, recall, and ROC-AUC. (Python, Pandas, Scikit-learn, TensorFlow)

TECHNICAL SKILLS

- **Computer Vision & ML:** YOLOv8 (Ultralytics), PyTorch, TensorFlow, OpenCV, ONNX export & benchmarking, Scikit-learn, dataset building & annotation (Roboflow), data augmentation
- **Full-Stack:** React 18, Node.js, Express, PostgreSQL, MongoDB, REST APIs, i18next (AR/EN RTL), Git/GitHub, WhatsApp bot integration
- **Data:** Python, Pandas, NumPy, EDA & data cleaning, Matplotlib, Seaborn, Plotly, Power BI, SQL
- **Embedded & Hardware:** ESP32, ESP32-CAM, Arduino, sensors, motor control, circuit design (Fritzing), power calculation, logic design
- **Workflow:** AI-assisted development (LLM tools), client requirements analysis, C, C++, Linux, Networking basics (CCNA 200-301)

EDUCATION

B.Sc. Computer Science — Faculty of Computers and Artificial Intelligence, Benha University

[2023] – [2027]

COURSES & CERTIFICATES

- Deep Learning — NVIDIA Deep Learning Institute (2025) • HCIA-AI V3.5 — Huawei (2025)
- Associate Data Scientist in Python — DataCamp • Understanding Data Science — DataCamp (2025)

- Prompt Engineering — Dubai Future Foundation (2025) • CCNA 200-301 (2023)

LANGUAGES

Arabic (native) • English (professional working proficiency)